



Simulation optimization approach for hybrid flow shop scheduling problem in semiconductor back-end manufacturing

James T. Lin^{*}, Chien-Ming Chen

National Tsing Hua University, No. 101, Section 2, Kuang-Fu Road, Hsinchu 30013, Taiwan

ARTICLE INFO

Article history:

Received 29 January 2013

Received in revised form 17 October 2014

Accepted 20 October 2014

Available online 19 December 2014

Keywords:

Simulation optimization

Genetic algorithm

Optimal computing budget allocation

Hybrid flow shop scheduling

Semiconductor back-end manufacturing

ABSTRACT

This study presents a simulation optimization approach for a hybrid flow shop scheduling problem in a real-world semiconductor back-end assembly facility. The complexity of the problem is determined based on demand and supply characteristics. Demand varies with orders characterized by different quantities, product types, and release times. Supply varies with the number of flexible manufacturing routes but is constrained in a multi-line/multi-stage production system that contains certain types and numbers of identical and unrelated parallel machines. An order is typically split into separate jobs for parallel processing and subsequently merged for completion to reduce flow time. Split jobs that apply the same qualified machine type per order are compiled for quality and traceability. The objective is to achieve the feasible minimal flow time by determining the optimal assignment of the production line and machine type at each stage for each order. A simulation optimization approach is adopted due to the complex and stochastic nature of the problem. The approach includes a simulation model for performance evaluation, an optimization strategy with application of a genetic algorithm, and an acceleration technique via an optimal computing budget allocation. Furthermore, scenario analyses of the different levels of demand, product mix, and lot sizing are performed to reveal the advantage of simulation. This study demonstrates the value of the simulation optimization approach for practical applications and provides directions for future research on the stochastic hybrid flow shop scheduling problem.

© 2014 Published by Elsevier B.V.

1. Introduction

The semiconductor manufacturing process can be divided into wafer fabrication, probing, assembly, and final testing steps. The back-end process of semiconductor manufacturing consists of the steps that follow wafer fabrication (front-end process). This study is motivated by a real-world back-end assembly scheduling problem observed at a leading outsourced semiconductor assembly and test (OSAT) company, which serves as a virtual factory for an integrated circuit (IC) design house (or fabless) or integrated device manufacturer (IDM). The OSAT adopts a typical make-to-order (MTO) and contract manufacturing model. The front-end facilities are advanced and complex from a technological point of view, whereas several problems arise in the back-end facilities from a logistics point of view [37].

Semiconductor back-end manufacturing usually operates with a short lead-time and absorbs the effect of lead-time variance in the front-end process. Two factors of influence should be considered in scheduling, namely, time to market

^{*} Corresponding author.

E-mail address: jtlin@ie.nthu.edu.tw (J.T. Lin).

and inventory control. At the initial stage of the product life cycle, back-end assembly provides customers with support by increasing profit and reducing sales loss. At the maturity stage, although a physical inventory is absent, shortage risks can be mitigated by quick responses. If the lead-time is short and robust, then the replenishment risk is relatively low because the inventory level is aligned with the actual demand. In contrast, a long lead-time corresponds to increased risks because the inventory level is based on the forecast with a bullwhip effect; both the potential inventory holding cost and sales loss are relatively high. Semiconductor back-end manufacturing has the aim of shortening lead-time via flow time reduction. This goal presents a challenge because of the complexity of a production system with limited capacity, particularly in the MTO and contract manufacturing model in which wafers are consigned by customers. The assembly facility cannot begin work on an order without the wafers even if capacity is sufficient. The complexities of scheduling problems in semiconductor back-end manufacturing arise from the wide product mix, large number of parallel machines, machine dedication, and numerous jobs in the production line [17,30,51]. Simulation and optimization are two of the most commonly used methods in operations research. In optimization, the best solution is selected using an exact procedure in mathematical form and under relatively strict assumptions. Heuristic or hybrid approaches are applied to obtain an approximate solution if an exact procedure is unavailable. An optimal solution may be infeasible if an exact solution is applied because of strict and deterministic assumptions. In contrast, traditional simulation describes the production system and defines the input, process, and output to predict system performance but lacks optimization capability. Simulation provides various what-if scenarios without physical implementation. This problem represents a real-world problem with a large solution space, such that applying an exact solution procedure to obtain optimal solution is difficult even if the optimization aspect is expected. Discrete optimization via simulation techniques has recently been developed to solve stochastic optimization problems with discrete decision variables. The barriers that hinder the wide application of this approach are the long computation time required for simulation and the noise of performance evaluation using simulation under stochastic conditions [27]. The aforementioned concerns must be overcome to arrive at a feasible procedure.

1.1. Problem statement

A typical back-end assembly operation consists of multiple stages, as shown in Fig. 1. The surface of the wafer is taped to avoid damage to the circuits (“Tape”). The thickness of the wafer is reduced to satisfy the packaging technology requirement (“Back Grind”). The wafer is fixed with film and a metal frame (“Detape/Mount”) and subsequently diced, and the scraps are washed out (“Saw”). The die is attached to a substrate or lead frame using adhesives (“Die Bond”). The die is linked to the substrate or lead frame using gold or copper wires, which also connect the die to the outer circuit (“Wire Bond”). The circuits of the components are moulded to protect them from damage by outside forces and to strengthen their physical characteristics (“Mould”). The next process is “Mark.” The chips are severed from the lead frame and lead fingers or from the substrate and balls are formed to connect the chips (“Trim/Form” or “Ball Placement”). The last step is “Final Visual” inspection. Upon completion, the product is transferred or shipped to testing facilities according to the customer’s instructions [54].

Multiple lines exist in an assembly facility. Line scale is normally measured (as a bottleneck) using the number of wire bonders (approximately ranging from 500 to 1000) to assess the efficiency of the management span. The number of other machines depends on the output of the wire bonding stage. Each stage contains different machine types, which are grouped with other machine types in parallel. Such an arrangement was established for two primary reasons: (1) the need for different machines to perform a process depending on the level of product quality or precision requirement and (2) the evolution of equipment technology or procurement strategies of different vendors. Thus, a multi-line/multi-stage production system with numbers of different identical and unrelated parallel machines was established.

Different customers require different product (device) types, and a single device is specified for each customer. Product engineers either initiate the product master data of a new product by means of a qualification procedure known as new product introduction (NPI) or modify the existing product master data by performing an engineering change management (ECM) process, i.e., adding a newly qualified machine type. After approval, the product master records the routing in a stage-by-stage manner, qualifies the production line(s), and qualifies the machine type(s) by recording the stage and counter-processing time separately for each product.

Orders continuously arrive in the MTO environment. The assembly facility commonly uses a bottleneck to schedule the order release time. However, a single bottleneck is insufficient because flexible routes and entities in the production behaviour are altered via splitting for parallel processing and merging for completion and shipment. Numerous orders with different release times are released daily. An order in back-end semiconductor manufacturing generally contains a specified

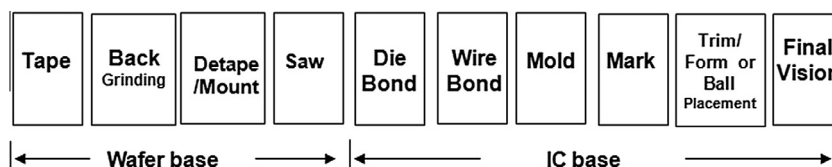


Fig. 1. Typical semiconductor backend assembly process.

contract customer, wafer lot number, device, and quantity. When an order is received at the factory, the first step is to select a production line from those available and select a machine type for each stage. The die bond, wire bond, and mould stages are critical in back-end manufacturing because these stages determine the flow time. Thus, this study focuses on these stages.

A quantity parameter splits the order. The value of this parameter represents the number of job movements across the stages. During NPI or ECM, the product engineer selects a substrate or lead frame that is qualified to serve as a carrier for the dies. Each substrate or lead frame contains a specific number of dies. Each magazine contains multiple strips of substrates or lead frames. The magazine is a container that handles job movement across stages; thus, the quantity in the magazine is the parameter used to split the order. This quantity is derived by multiplying the number of dies in the strip with the number of strips in the magazine.

Fig. 2 illustrates the example of the production environment and entity (order and job) flow as production behaviour. Two production lines with three stages exist in alternative 1. Each stage contains different types of machine with different numbers. For example, two types of wire bonders (WB1 and WB2) and three machines exist in production line 1, whereas only one type of wire bonder (WB2) and four machines exist in production line 2. Three orders are present in this example. Order 2 (big black dot) has a specific product type and demand quantity. The order is assigned to a qualified die bonder (e.g., DB1). Fig. 3 shows the detailed process at the die bond stage, from picking the die from the wafers to placing the substrate into the magazine. This process is repeated until the quantity requirement of one magazine is satisfied. The order is split into jobs (small black dots) for the wire bond stage. Again, one machine type is selected among the wire bonders (e.g., WB1). The job moves to the mould stage, which is contained in the magazine. One machine type is selected among the moulding machines (e.g., MD1). Thus, all of the jobs split from the order undergo simultaneous parallel processing, and they all use the same machine type at each stage and are merged at the end after completion. The flow time of a single order is calculated from the release time of the order until all subsequent jobs are completed. Supervisors experience difficulty in managing bottleneck shifting issues and production behaviours in order splitting and merging because of multiple orders and jobs

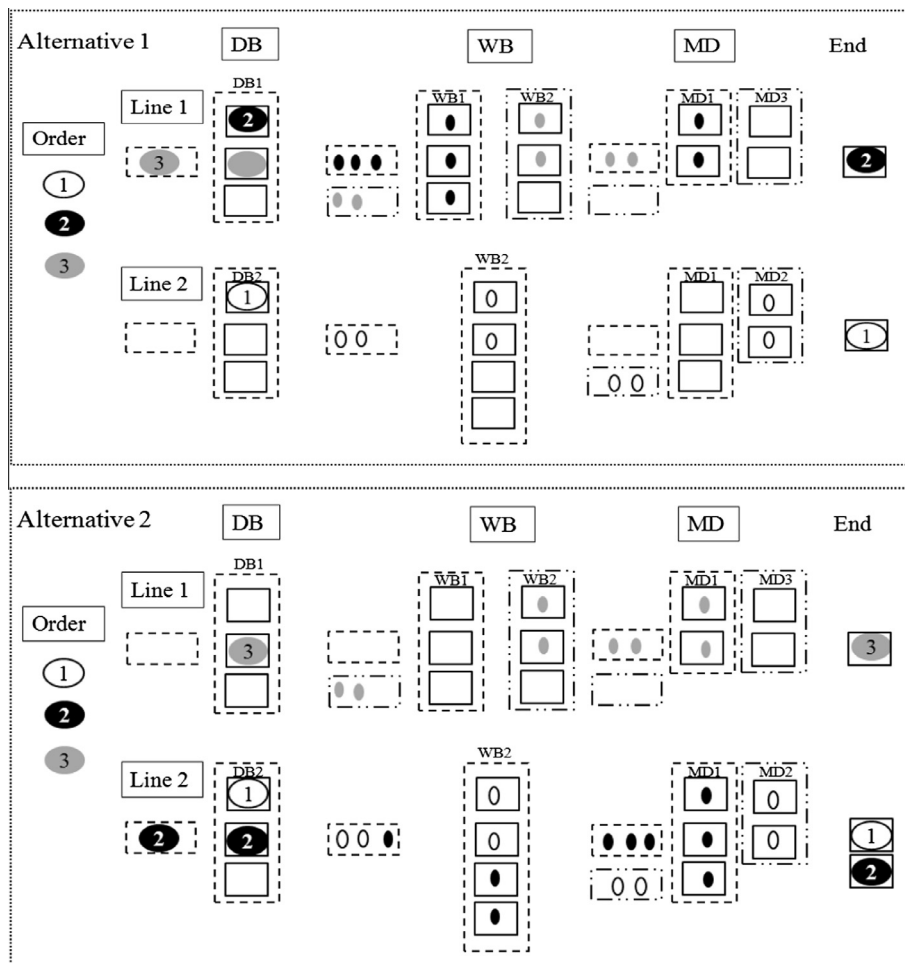


Fig. 2. Production line and examples of assignment and production behaviour.

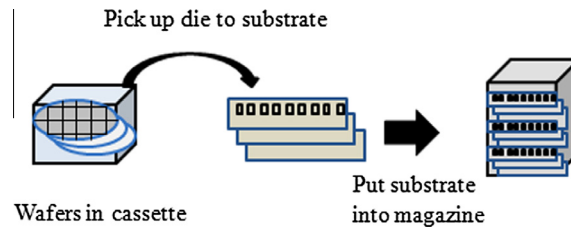


Fig. 3. Detailed process at the die bond stage.

in the production system. Alternatives 1 and 2 in Fig. 2 are examples of different assignments of production lines and machine types per stage and order. Different assignments will have different flow time performances because of the differences in processing time, waiting time in the buffer, and setup time, particularly when the device is changed during stage movements. Fig. 2 is only an example; other alternatives also can be used. Complexity increases if the orders are increased.

Complexity arises because each order has several alternatives for route options. Such alternatives are constrained by the capacity, capability, and production conditions of the system. Multiple orders arrive with varying demands for devices, specific customers, wafer lot numbers, and quantities. The capacity constraint indicates that a limited number of parallel machines exist in the production system. The capability constraint indicates that each product can only be produced on a qualified line(s) and machine type(s) with the countering different stochastic processing time. The production behaviour conditions denote the following: (1) order splitting and merging during production, (2) sequence-dependent setup times that occur once a product change occurs during processing in a machine, and (3) the requirement that all jobs of the same order use the same machine type at each stage.

This study focuses on the assignment of orders and their subsequent jobs to a particular production line and machine type to achieve minimal flow time subject to the capacity, capability, and production behaviour conditions. The problem is an extension of the hybrid flow shop scheduling problem (HFSSP). However, this study differs from those on traditional HFSSP for the following reasons: (1) extension from single to multiple lines, (2) assumption of parallel machines from simply identical or unrelated machines to groups of various machines, and (3) production behaviour under specific conditions.

The remainder of the paper is organized as follows. Section 2 discusses the related literature on similar problems in practical and theoretical areas. The simulation optimization approach is introduced in Section 3, and this section also provides a discussion of the simulation moulding and optimization steps. Section 4 illustrates an empirical study, including parameter justification and the differences in performance resulting from the use of different objective perspectives and techniques; a scenario analysis is also presented in this section. Section 5 provides the conclusions.

2. Literature review

2.1. Related studies on scheduling problems in semiconductor back-end manufacturing

A number of capacity planning and scheduling characteristics are apparent in semiconductor back-end manufacturing. These characteristics are generally divided into two categories, namely, demand from monthly or weekly forecasts and capacity planning focused on product mix determination based on available capacity and tools. This classification provides guidance for capacity allocation. The demand from actual orders undergoes execution-level capacity planning, problem scheduling focused on machine assignment, flow time reduction, and shipping date establishment. A mathematical model is usually adopted to forecast the demand. Sweat proposed a mixed-integer linear programming model to address product mix planning and capacity allocation in a multi-factory environment [45]. Wang et al. presented an alignment based on a forecast for mid-term capacity planning and a scheduling framework to address the discrepancy between scheduling and machine allocation for orders in a back-end assembly environment [50]. Potoradi et al. employed simulation-based scheduling to maximize demand fulfilment. Discrete event simulation was used to reduce tardiness by reducing the setup time and prioritizing lots in the lot release mechanism. Product diversity, the use of wire bonders as bottleneck machine groups, and sub-lot parallel processing in the assembly facility were also discussed. However, an optimization scheme has yet to be added [36].

Tang et al. proposed a heuristic algorithm to configure a large system into a virtual production line design to ensure line balance and efficiency; such an approach was demonstrated in a simplified back-end semiconductor manufacturing system [46]. Pearn et al. considered parallel machine scheduling (PMS) on the die bonder using integer programming for scheduling with respect to machine capability in terms of product type and setup time. Complexity was shown to be non-deterministic polynomial-time (NP)-hard and was resolved by a heuristic algorithm [32]. Yang and Shen reported that wire bonding was a bottleneck and proposed a heuristic algorithm to reduce completion time [52]. Chua et al. presented an intelligent multi-constraint finite-capacity-based lot release system to study the lot release mechanism [15]. These studies also examined the complexity in large product mixes and multitudes of capacity constraints [29,53]. Jarugumilli et al. proposed a framework for execution-level capacity allocation for semiconductor back-end manufacturing through the use of a compromised

hybrid optimization–simulation model; however, the simulation was simply used for checking feasibility and not for optimization [20]. Chen et al. presented integrated modules for an infinite capacity planning system in lot release and finish time estimation, and both capacity and capability properties were considered. Various types of machines with a tool configuration relationship have been proposed. Each product is treated in accordance with equipment capability. The abovementioned studies included work-in-process (WIP) pulling, lot release, resource selection, workload accumulation, and workload balance modules [13].

Sivakumar and Gupta developed a Pareto optimal scheduling technique through simulation to determine the job sequence for the probing process, and a trade-off was implemented between cycle time and utilization [42,43]. Weigert et al. applied a heuristic method in simulation-based scheduling to optimize the operation sequence; the cycle time, throughput, and due date compliance were considered in this method. In the study, the disadvantages of the simulation approach and long simulation time were discussed, and a method was proposed to overcome the disadvantages [51]. Klemmt and Weigert developed a hybrid probing approach by considering such complexities as machine dedication, setup, and processing time dependence. The allocation result was applied in the optimal capacity allocation simulation, and the production behaviour was considered to achieve exact scheduling. After the interactions between the exact and simulation approaches, the setup times and the number of setups were reduced [22].

A review of the related literature reveals that the complexity of semiconductor back-end manufacturing arises from the variety of product mixes, bottleneck considerations, parallel machines, and available routings. Recent studies have employed the simulation approach alone or have integrated it with heuristic algorithms because of the complexity.

2.2. HFSSP with parallel machines

Two particular types of scheduling problems exist in HFSSP. These problems are PMS and flow shop scheduling (FSS). The PMS refers to the allocation of jobs to machines, and FSS refers to the sequence of jobs. Ribas et al. recently reviewed published papers on PMS and FSS and classified the HFSSP problem in terms of production system and solution procedure. A standard hybrid flow shop structure is shown in Fig. 4. The structure is composed of more than one stage in the sequence flow and at least one stage that includes more than one machine. Each machine in each stage can process one job at a time. Common assumptions in HFSSP include the following: all jobs and machines are available at time zero, the machines are identical, setup times are negligible, pre-emption is not allowed, buffer size is unlimited, and problems are deterministic. In reality, these assumptions are inconsistent with practical scenarios or academic concerns, and hence, further studies have been performed to analyse HFSSP [38–40]. The HFSSP is generally an NP-hard and complex combinatorial problem encountered in numerous actual situations. Scheduling problems with parallel machines and sequence-dependent setup times are difficult to solve. The single-machine scheduling problem with sequence-dependent setup times is known to be NP-hard [35]. For the parallel machine case, the problem of minimizing the make-span with two identical machines is NP-hard [18]. Thus, the more complex case of minimizing the make-span on a scheduling problem with identical parallel machines and sequence-dependent setup times is also NP-hard [23,31].

Production systems vary in terms of machines, job constraints, and objectives, and machines are classified as identical, uniform, or unrelated. Complexities are primarily caused by job constraints. Non-hybrid specific constraints include setup time, processing time, and job precedence, whereas hybrid specific constraints include re-entry, lot-size scheduling, lot splits, and other factors. The objectives are classified as WIP, utilization, idle time, orders served, flow time, and inventory. The solutions vary in terms of exact, heuristic, hybrid, or simulation/decision support systems. The heuristics group is divided into constructive and improvement procedures or meta-heuristics [38]. Sun et al. reviewed a multi-objective optimization algorithm for a large and complex FSS problem [44]. Chaari et al. adopted a genetic algorithm (GA) for a robust HFSSP under demand uncertainty [7]. Chaudhry and Drake applied GA for machine scheduling and worker assignment to minimize total tardiness [9]. Chan et al. presented a GA-based job shop scheduler for a flexible, multi-product, parallel-machine sheet-metal job shop [8]. Almeder and Hartl applied a variable neighbourhood search-based simulation optimization approach to a stochastic flexible flow shop problem. Simulation optimization outperforms deterministic exact procedures in terms of considering stochastic and detailed production behaviour. Moreover, simulation optimization considers the utilization of machines, buffer, and cycle time [1].

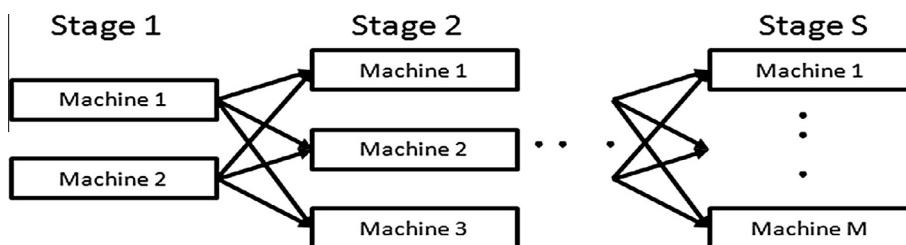


Fig. 4. Hybrid flow shop standard structure.

The use of completion time to measure performance in scheduling problems is subject to general criteria. The minimization of F_{max} (the maximum flow time) essentially indicates that the cost of a schedule is directly related to the job with the longest duration. The minimization of C_{max} (the maximum completion time) suggests that the cost of a schedule depends on the overall time required by the processing system to complete the entire set of jobs. The value of C_{max} and F_{max} are identical if the ready times are zero. However, these variables differ significantly when the ready times are non-zero. The value of $\bar{F}$ (the mean flow time) is minimized through $\frac{1}{n} \sum_{j=1}^n F_j$, where the flow time F_j of an order j is defined as $F_j = C_j - r_j$. This expression denotes the difference between the completion and release times of order j [16]. The value of $\bar{F}$ is derived from total flow time, which may result in starvation, i.e., certain jobs may be delayed to an unbound extent. To address this limitation, F_{max} is considered as well [2].

The HFSSP has elicited considerable research attention in the last 50 years. Actual production systems vary, and thus, their specific characteristics and suitable procedures should be considered in their solutions. To our knowledge, only a few studies have considered multi-line and group parallel machines, including identical and unrelated machines as well as simultaneous considerations of specific production conditions under a stochastic situation. Moreover, only a few studies have been published on the use of the simulation optimization approach [38].

2.3. Simulation optimization and optimal computing budget allocation

Carson and Maria defined simulation optimization as the process of identifying the best input variable values without explicit evaluation. A simulation optimization model is shown in Fig. 5. The model is a loop created by simulation and optimization [6]. According to Tekin and Sabuncuoglu, the most accurate systems are highly complex. Such a level of complexity renders performance computation and analytical identification of optimal decision variables difficult and at times impossible. Simulation optimization is a powerful decision-making tool that is capable of capturing the intricate relationships and interactions among various entities in a real-world complex system and subsequently identifying the best design or solution [28]. The problems in simulation optimization can be categorized into local and global categories. Local optimization techniques are further classified in terms of discrete and continuous parameters. Search algorithms for global optimization include evolutionary algorithms (EAs), tabu search, simulated annealing, Bayesian/sampling algorithms, and the gradient surface method [47]. Branke and Mattfeld applied an EA to a dynamic scheduling problem [4]. Recently extended studies used EAs to establish optimal dispatching rules [3]. The GA is one of the most popular and widely applied EAs for scheduling problems [33,48].

Stochastic simulation is a powerful modelling and software tool for analysing modern complex systems prior to physical implementation, and this method can accurately specify a system with logically complex and non-algebraic variables and constraints [25]. However, the definition of system behaviour from the input value and that of the process from the output value are critical factors. Practitioners should understand the system performance and key factors from a what-if perspective, which is a challenge because no optimal solution can be obtained from simulations. Simulation optimization is considered an alternative solution to this problem.

Simulation optimization refers to stochastic optimization using simulation techniques, such as Monte Carlo simulation or discrete-event simulation. However, system behaviour is stochastic. Replications that consider statistics are needed to measure the output, and such a procedure consumes a substantial amount of simulation resources. According to the law of large numbers, the larger the sample size, the more likely it is that values obtained from the sample will be similar to the actual values for the population. Many researchers use a sample size of 30 as a goal in planning research [19]. If k alternatives are observed, we obtain $k \times 30$ replications. Chen et al. proposed optimal computing budget allocation (OCBA) as an ultimate simulation resource [11,12]. The OCBA is one of the three main approaches used to solve the selection problem. The other two approaches are the indifference zone and the expected value of information procedure [5]. The OCBA applies statistics rather than the equal rule. The n_0 replication times are initially necessary for each selected alternative to obtain the first estimation and to subsequently determine the differences among the alternatives in later steps. Thus, simulation resources are only allocated by delta (Δ) times (replications) to similar performances with large variance until the statistical probability of the correct selection (PCS or $P\{CS\}$) satisfies the target or until the simulation resource is exhausted. The OCBA maximizes the advantage of simulation optimization because it can minimize the number of replications in stochastic systems [10,12]. Schmidt et al. proposed the idea of integrating OCBA and EAs. The goal was to drastically reduce the number of evaluations required for proper operation of the evolutionary algorithm in noisy environments [41]. The OCBA has two applications. The first application is the direct reduction of replications in local optimization to determine the best of a finite set of

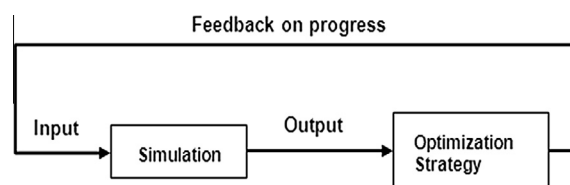


Fig. 5. Simulation optimization model.

alternatives. The second application is indirect integration with search algorithms, which should repeatedly evaluate and compare potential solutions during the search process to determine the subsequent search direction [24]. Simulation optimization continues to be applied and developed to solve complex problems [25].

3. Proposed approach

We consider an NP-hard hybrid flow shop scheduling problem. Decisions are discrete for order assignment relative to production line and machine type with specific conditions, i.e., product-machine dedication, lot split, and merge. However, jobs that belong to the same order must be processed by the same machine type at each stage. Processing time is the stochastic cause of machine availability and forms a stochastic discrete optimization problem subject to non-closed-form constraints. Simulation optimization is a typical solution employed for this type of problem. Given that a large solution space and a simulation approach are used, we couple a meta-heuristic and acceleration technique based on GA and OCBA to address the issues related to the simulation budget and estimation errors. This study subsequently demonstrates the advantage of the simulation approach.

This section defines basic terminology to facilitate understanding of the production system. Simulation modelling is subsequently introduced as an evaluation mechanism. Based on the problem objective and the selected optimization strategy by GA, the decision variables are designed and embedded into the simulation model. Finally, the steps of simulation optimization approach are detailed in a flow chart.

3.1. Terminology

The basic terms that describe the production layout include production line, stage, machine and product types, order attributes, jobs behaviours, and performance measurement.

Production line: A line that contains the required machine types in a production stage for order completion. Multiple production lines exist in a production system.

Production stage: A stage with the same process function. Multiple parallel machines are classified by their types. A number of stages are present in a production line.

Machine type: Refers to a specific machine based on the demand by the vendor or the capability level for product precision. Various machine types are present in each stage of a production line.

Product (device) type: Refers to specific device, a qualified production line and machine type, and the counter-stochastic processing time cause of machine availability.

Order: Refers to the specific customer demand for the device and quantity with a specific scheduled release time.

Job: A job is split from an order, the quantity of which is defined as the volume of one magazine. Jobs are simultaneously processed in parallel. Every job with the same order can only be processed by the same machine type for each stage. No material shortage occurs during production.

Setup time: Setup time is established if the product type between orders or jobs (current to next) is changed in an assigned machine. Setup time varies by stage.

Performance measurement: Under MTO, all orders with different release times must be completed within a minimal flow time; thus, we measure performance using $\bar{F}$ and F_{max} . Both values are minimized in the current study.

3.2. Simulation modelling

Plant Simulation (formerly eM-Plant) was applied in this research to construct the production system for order and job processing. A simulation tool generally offers several advantages. An object-oriented simulation design is helpful in configuring a production system. Generic objects, such as icons, are used to describe or define the system inputs, process, outputs (including the physical production line, buffers, and process centres), order, and entity flow with the control rule and data collection mechanism for the required event records (Fig. 6). An object-oriented design allows the user to set up a number of parameters, i.e., time, quantity, and distribution, as attributes for each entity (order and job) and its relationships (jobs are the consequence of an order), buffer, or process centre. Such design also allows for the easy construction of a production system. A simulation tool can also represent the level of detailed architecture. Various pre-defined functions can aid users in gathering statistics for further analysis. An open function is not required to be integrated with another system or design for a number of algorithms. Plant Simulation provides “simtalk.” Thus, this method can be used to bundle the optimization design.

We designed and observed the production layout and behaviour, constructed a simulation model, and defined the flow based on production behaviour. The basic steps in the simulation model consist of identifying the object, designing the level of architecture, and linking the relationships among the objects. Once the system is built, model verification and validation were performed to ensure that the simulation model is accurate and suitable for representing the actual production system. Verification was performed from a technical perspective to verify whether the model is well transformed. Validation was

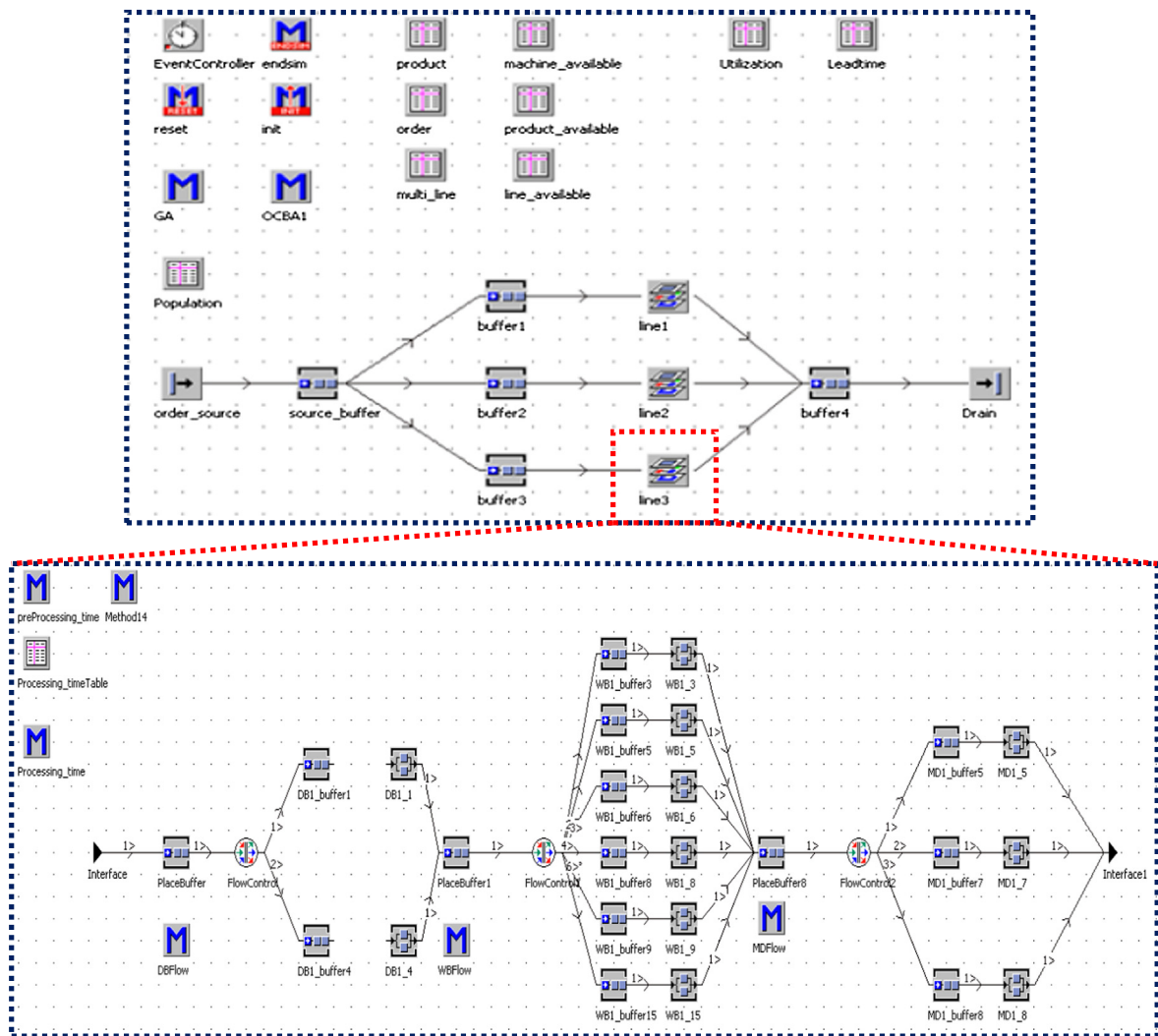


Fig. 6. Simulation modelling.

based on system content to ensure that the model represents the production system. Data validation methods used by experienced managers in a real production system were adopted.

3.3. GA encoding for the decision variable

Two decisions are required in this problem. The first decision assigns the orders to the production line, and the second assigns the jobs to the machine types for each stage. Thus, the chromosome structure is used to represent the decision variables. The first component is order assignment. For example, if three orders are received, then three genes are defined to represent the assigned production line. The second component is machine assignment for each stage and order. For example, if process routing has three stages, then three genes are designed to represent the machine assignment. Fig. 7 shows the examples and reveals the complexity of the problem. If each order is produced at lines 1 and 2, then each order is allowed

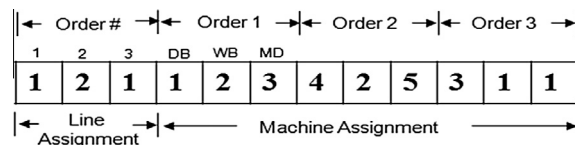


Fig. 7. GA encoding structure.

upper bond T times as the termination condition in limited-resource simulation. The GA is a search algorithm, whereas OCBA is a simulation accelerator. The GA and OCBA loops continue until the termination condition is satisfied. In summary, GA functions as a search algorithm from the global view, and OCBA effectively ranks and selects better solutions in the population from the local view. The interactions between GA and OCBA can assist in identifying an optimal solution through reduced replication, thereby overcoming the need for long replications during simulation.

4. Illustration

4.1. Empirical study

This study used a set of data obtained from an assembly facility. The problem description includes the facility layout, profile of the product (device), and orders and involves three production lines and three stages. The data include five types with 84 machines at the die bond stage, 15 types with 814 machines at the wire bond stage, 10 types with 47 moulding systems at the mould stage, and 60 types in the product file. Each product contains counter-qualified production and machine types with stochastic processing time distributions. The setup time for each product was also defined during product change. A total of 105 orders exist, each with a specific product, demand quantity, and scheduled start time. The simulation model collects the flow time from order arrival to completion time based on the different assignments and utilization of each machine type at each stage and production line. Search algorithms and OCBA are important in applying the simulation optimization approach.

The GA parameters, namely, crossover rate (CR) and mutation rate (MR), were considered because the algorithm is related to solution effectiveness or quality. The OCBA parameters were subsequently considered because the strategy is related to replication reduction. Two levels were set for each GA parameter: 0.6 and 0.8 for the crossover rate and 0.1 and 0.2 for the mutation rate. The ANOVA results are shown in Table 1. The crossover rate is the main factor used to assess performance with a statistically significant difference. The interaction results for $\bar{F}$ and F_{max} are shown in Fig. 9. The results show that the suitable levels for the crossover and mutation rate are 0.8 and 0.1, respectively.

The OCBA parameters, i.e., n_0 , delta, and $P\{CS\}$, were tested to verify the replication level as well as the individual effects and interactions under different parameter settings for later optimization steps. Intuitively, larger parameters result in more replications, whereas the alternatives are the same in the OCBA procedure. However, this test involves a different combination of parameters, and the designs (alternatives in the population) are different or changed because of the evolutionary

Table 1
ANOVA for the performance of $\bar{F}$ and F_{max} .

Source	DF	P-value ^a	
		$\bar{F}$	F_{max}
CR	1	0.018 [*]	0.013 [*]
MR	1	0.22	0.198
CR*MR	1	0.204	0.663
Error	4	–	–
Total	7	–	–

$\bar{F}$: $S = 7906.29$, $R\text{-Sq} = 82.84\%$, $R\text{-Sq (adj)} = 69.97\%$.

F_{max} : $S = 19136.2$, $R\text{-Sq} = 83.61\%$, $R\text{-Sq (adj)} = 71.31\%$.

^a $\alpha \leq 0.05$.

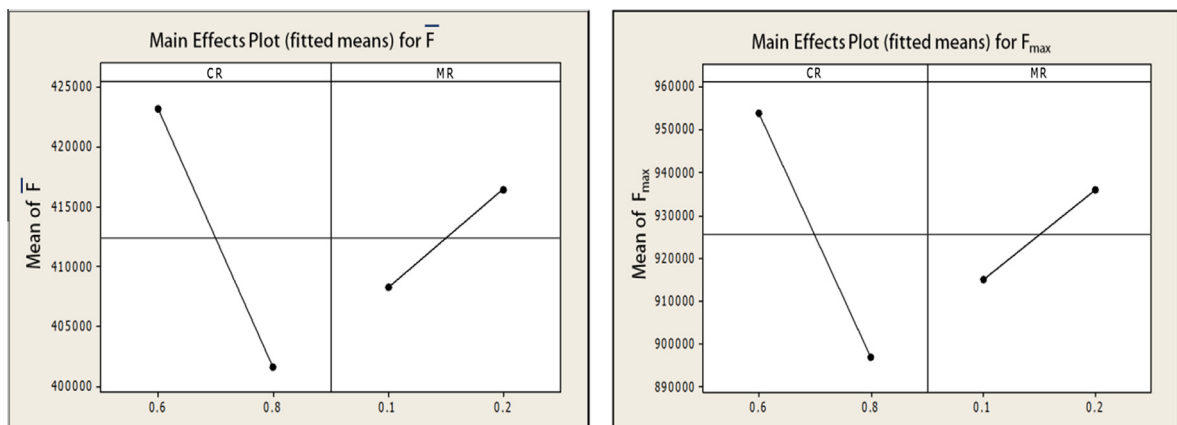


Fig. 9. Main effect of GA parameters, $\bar{F}$ and F_{max} .

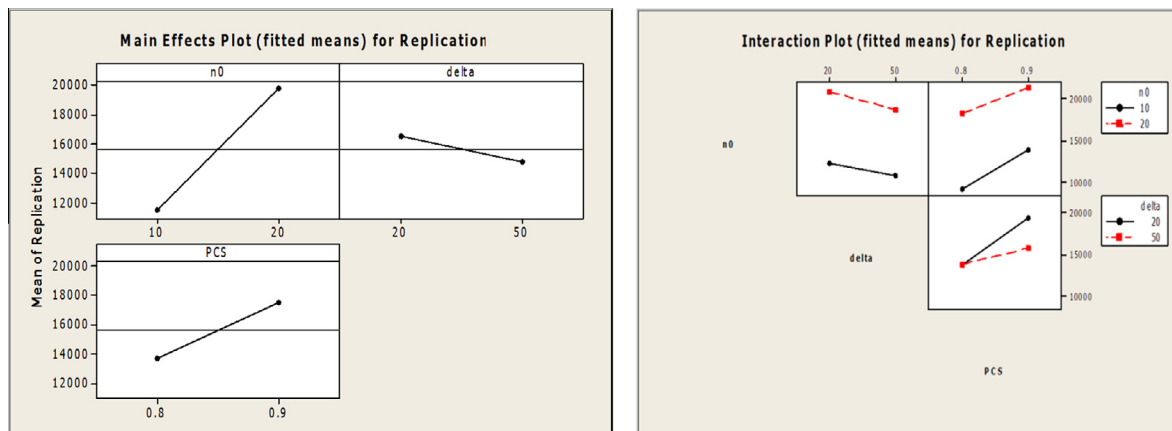
procedure applied by GA over 30 generations. Two levels were set for each parameter: 10 and 20 for n_0 , 20 and 50 for delta, and 0.8 and 0.9 for $P\{CS\}$. The ANOVA, $\bar{F}$, and F_{max} results are shown in Table 2. The test result for $\bar{F}$ shows that n_0 , delta, $P\{CS\}$, and the interaction between delta and $P\{CS\}$ are the main replication factors. The results for F_{max} show that n_0 , $P\{CS\}$, and the interaction between n_0 and $P\{CS\}$ are the main replication factors. The observed result of delta for $\bar{F}$ revealed that large delta (50) incurred fewer total replications than small delta (20), and for F_{max} , the result revealed that delta is not significant to total replications. This phenomenon may be caused by different selected designs during the evolutionary cycle and not only by delta, or the effect of delta may be relatively smaller than other parameters. By default, T is 900 per iteration, which follows the supposed maximum of 30 times for each alternative and the 30 total alternatives in a population. As shown in Figs. 10 and 11, the suitable values for n_0 , delta, and $P\{CS\}$ are 10, 50, and 0.8, respectively.

The proposed simulation optimization approach can be performed based on the suitable parameters of GA and OCBA. We adopt GA as an optimization strategy and combine it with OCBA (noted as GA_{OCBA}) and traditional estimation in which 30 replications are performed equally for each solution (noted as GA_{Equal}) for further comparison. The termination condition of evolution consists of a stop after the 30th generation from both the $\bar{F}$ and F_{max} performance perspectives. Hence, we use this base to yield 27,000 replications (30 alternatives in population $\times$ 30 replications $\times$ 30 generations) from GA_{Equal} . The number of replications indicates the difference between GA_{OCBA} and GA_{Equal} , as shown in Fig. 12. The convergence behaviour shows the trends for obtaining the optimal solution, as shown in Fig. 13. Randomness and gene quality are crucial for convergence behaviour, and in this study, they revealed that GA_{OCBA} outperforms the traditional GA_{Equal} approach in terms of evolutionary computations at different flow time measurements. The improved performance cannot be solely attributed to OCBA because the search algorithms are similar if GA is used. The GA_{OCBA} continues to improve the performance until the 21st generation from $\bar{F}$ and until the 24th generation from the F_{max} perspective. Thus, if the simulation resource (time) is the same for both GA_{Equal} and GA_{OCBA} , improved performance can be attained because GA combined with OCBA can perform numerous generations in one gene revolution; in addition, OCBA significantly uses fewer replications. The GA_{OCBA} requires approximately 25 min (against the maximum allowed time of 30 min) to obtain solutions using an Intel Core i3 CPU with 3.3 GHz and 3 G RAM.

Table 2

ANOVA for the replications of $\bar{F}$ and F_{max} .

Source	DF	P-value*	
		$\bar{F}$	F_{max}
n_0	1	0.009*	0.01*
delta	1	0.042*	0.213
PCS	1	0.02*	0.024*
n_0 *delta	1	0.192	0.812
delta*PCS	1	0.04*	0.105
n_0 *PCS	1	0.097	0.041*
Error	1	–	–
Total	7	–	–

 $\bar{F}$: $S = 166.170$, $R\text{-Sq} = 99.98\%$, $R\text{-Sq}(\text{adj}) = 99.89\%$. F_{max} : $S = 162.635$, $R\text{-Sq} = 99.98\%$, $R\text{-Sq}(\text{adj}) = 99.87\%$.* $\alpha \leq 0.05$.Fig. 10. Main effect and interaction of OCBA parameters in terms of $\bar{F}$.

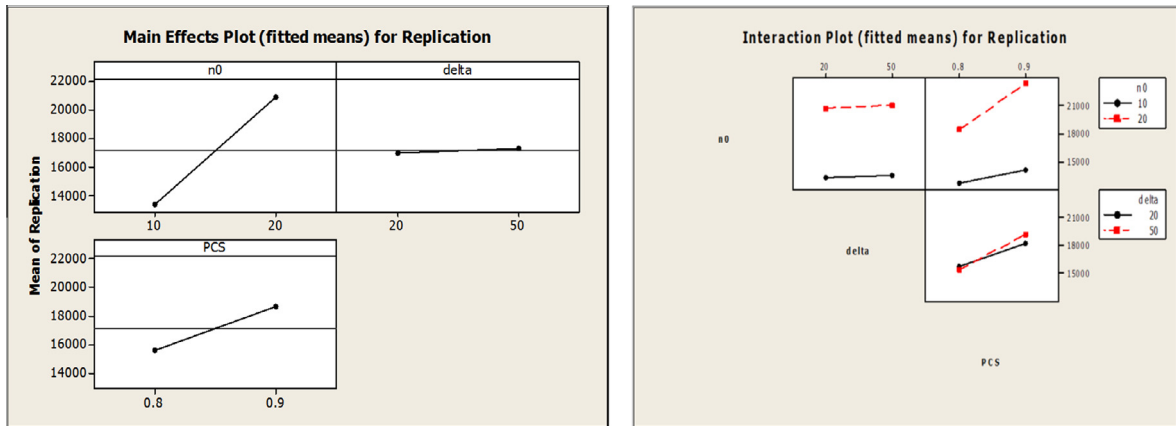
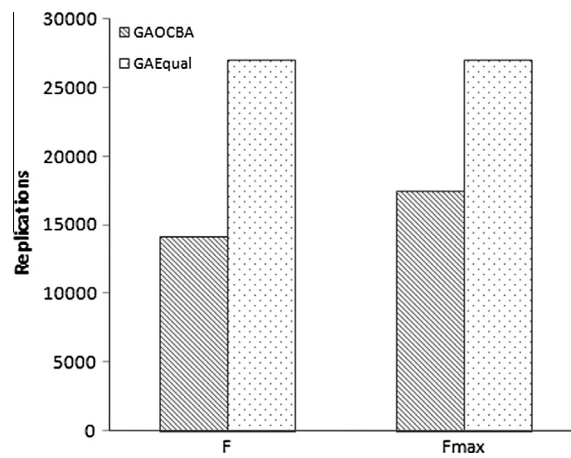
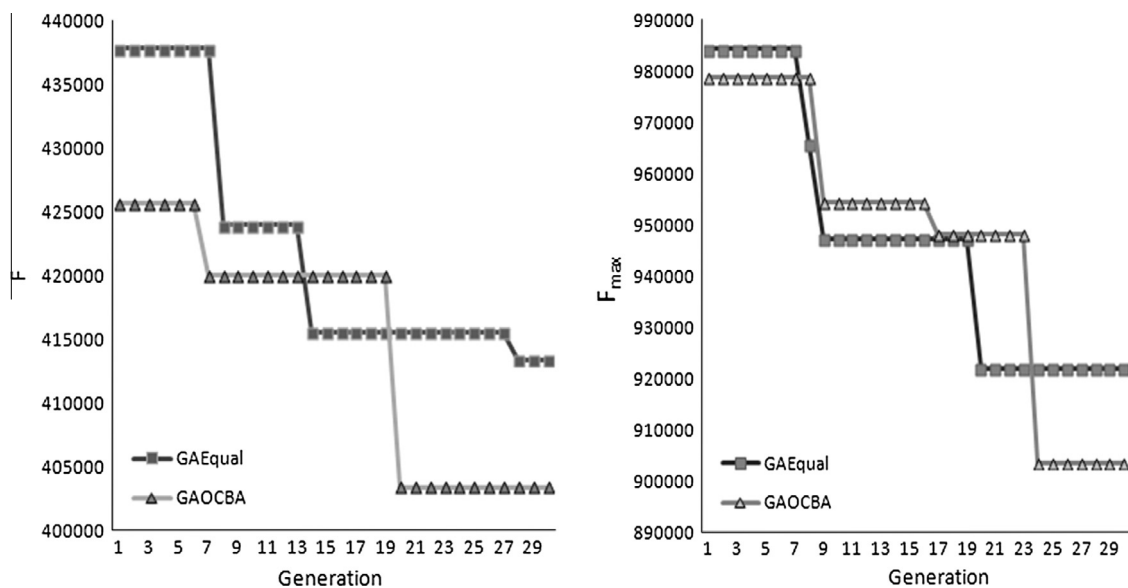
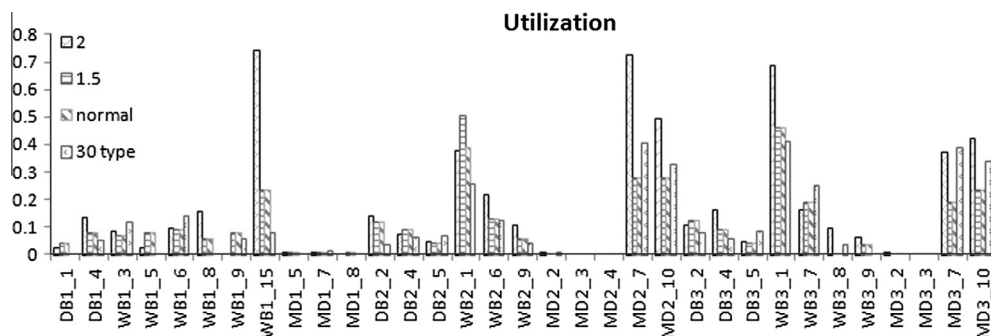
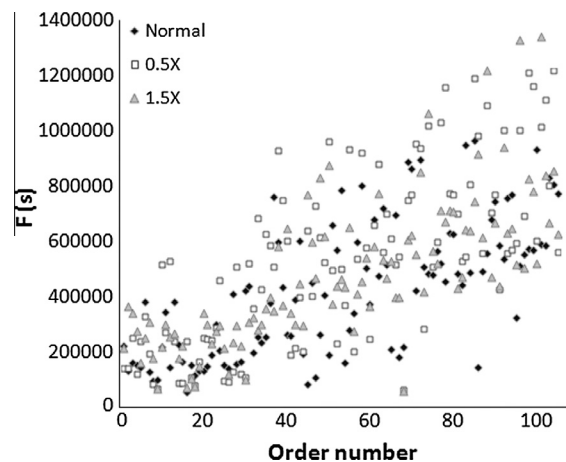
Fig. 11. Main effect and interaction of OCBA parameters in terms of F_{max} .Fig. 12. Comparison of replications with GA_{OCBA} and GA_{Equal}.

Fig. 13. Obtained performance and convergence behaviour.

Table 3Comparison of the performance of the different scenarios in terms of $\bar{F}$ and F_{max} .

Flow time performance ^a	Current demand/60 product types	1.5 × demand/60 product types	2 × demand/60 product types	Current demand/30 product types
$\bar{F}$	405,596	593,892	704,246	347,955
F_{max}	915,027	1,186,420	1,385,449	772,679

^a In second.**Fig. 14.** Utilization status among different scenarios.**Fig. 15.** Different performance by different lot sizes for the same order.

4.2. Scenario analysis

Practitioners in MTO and contract manufacturing are interested in the relationship between utilization and flow time performance for different demand levels and product mixes. Table 3 shows that flow time increases with demand but decreases with a decrease in the number of product mixes. Such phenomenon is caused by the saved setup time. Fig. 14 shows that utilization is one of the indicators for bottleneck prediction and awareness. This information aids practitioners in triggering ECM to release potential bottlenecks. Certain machine types maintain a high level of utilization, whereas others maintain a low level, and engineers can trigger these via qualified and share loading machines.

As an aside, a job is split from an order and the quantity in a job is fixed in our original problem, but practitioners expect to reduce flow time using a different quantity. In general, practitioners expect that a smaller quantity in a job (e.g., 0.5×) can reduce the flow time because of parallel processing. As illustrated in Fig. 15, we apply an instance to show the same order with a different splitting level (Normal, 0.5× and 1.5×), and we demonstrate its F performance. We encounter difficulty in stating which level is best because the result of each order varies. Certain orders are better, whereas others are worse. This issue should be studied relative to lot streaming in the hybrid flow shop scheduling problem, which considers splitting numbers as a decision variable for shortening flow time [14,21,49].

5. Conclusion

Semiconductor back-end manufacturing employs bottleneck scheduling through a lot release mechanism such that the ship-out dates can be easily estimated. However, this mechanism disregards bottleneck shifting from the PMS level while providing a space to improve flow time performance. Optimization is an opaque black box from the perspective of a practitioner. However, formulating a mathematical model and implementing an exact solution are difficult from the viewpoint of an academician because the problem in question is NP-hard and involves stochastic discrete optimization subject to non-closed-form constraints. In this study, simulation optimization coupled with GA and OCBA is used within an acceptable computation time. This approach aids in assigning orders optimally to the proper production line and machine types while achieving minimal flow time. This approach encourages experienced practitioners and provides a win-win situation for production planners, manufacturing operators, and customers. Additionally, the advantage of the simulation approach is maximized through various scenario analyses with managerial implications. One example shows the relationships between utilization and flow time under the condition of demand or product mix changes such that further studies may consider utilization as a multi-objective problem. Another example is lot sizing, which can facilitate the improvement of flow time through optimized lot sizing. These subjects are extended topics in HFSSP research. The proposed approach also provides an opportunity to integrate data from an actual production system and develop a decision support system through the use of simulation optimization, which is a highly useful strategy for dynamic production environments.

Subsequent studies could employ different combinations of other search algorithms and acceleration techniques to improve solution quality while adopting the simulation optimization approach. With respect to industrial applications with different objectives (i.e., WIP, tardiness, and utilization), further studies could analyse and design different production systems with specific characteristics or production behaviours.

Acknowledgements

The authors thank the referee for insightful comments that have helped to improve this work. The work in this paper was supported by NSC, Taiwan, R.O.C. (Project number: 101-2221-E-007-066-MY3).

References

- [1] C. Almeder, R.F. Hartl, A metaheuristic optimization approach for a real-world stochastic flexible flow shop problem with limited buffer, *Int. J. Prod. Econ.* 145 (1) (2013) 88–95.
- [2] C. Ambühl, M. Mastrolilli, On-line scheduling to minimize max flow time: an optimal preemptive algorithm, *Oper. Res. Lett.* 33 (6) (2005) 597–602.
- [3] J. Branke, C.W. Pickardt, Evolutionary search for difficult problem instances to support the design of job shop dispatching rules, *Eur. J. Oper. Res.* 212 (1) (2011) 22–32.
- [4] J. Branke, D.C. Mattfeld, Anticipation and flexibility in dynamic scheduling, *Int. J. Prod. Res.* 43 (15) (2005) 3103–3129.
- [5] J. Branke, S.E. Chick, C. Schmidt, Selecting a selection procedure, *Manage. Sci.* 53 (12) (2007) 1916–1932.
- [6] Y. Carson, A. Maria, Simulation optimization: methods and applications, in: *Proceedings of the 1997 Winter Simulation Conference*, 1997, pp. 118–122.
- [7] T. Chaari, S. Chaabane, T. Loukil, D. Trentesaux, A genetic algorithm for robust hybrid flow shop scheduling, *Int. J. Comput. Integr. Manuf.* 24 (9) (2011) 821–833.
- [8] F.T.S. Chan, K.L. Choy, Bibhushan, A genetic-based scheduler for multiproduct parallel machine sheet metal job shop, *Expert Syst. Appl.* 38 (2011) 8703–8715.
- [9] I.A. Chaudhry, P.R. Drake, Minimizing total tardiness for the machine scheduling and worker assignment problems in identical parallel machines using genetic algorithms, *Int. J. Adv. Manuf. Technol.* 42 (2009) 581–594.
- [10] C.H. Chen, K. Donohue, E. Yucesan, J. Lin, Optimal computing budget allocation for Monte Carlo simulation with application to product design, *Simul. Model. Pract. Theory* 11 (2003) 57–74.
- [11] C.H. Chen, L.H. Lee, *Stochastic Simulation Optimization – An Optimal Computing Budget Allocation*, World Scientific Publishing Co. Ptd. Ltd., Singapore, 2011.
- [12] C.H. Chen, J. Lin, E. Yucesan, S.E. Chick, Simulation budget allocation for further enhancing the efficiency of ordinal optimization, *Discrete Event Dyn. Syst.* 10 (2000) 251–270.
- [13] J.C. Chen, L. Su, C. Sun, M. Hsu, Infinite capacity planning for IC packaging plants, *Int. J. Prod. Res.* 48 (19) (2010) 5729–5748.
- [14] M. Cheng, N.J. Mukherjee, S.C. Sarin, A review of lot streaming, *Int. J. Prod. Res.* 51 (23–24) (2013) 7023–7045.
- [15] T.J. Chua, M. Liu, F. Wang, W. Yan, T. Cai, An intelligent multi-constraint finite capacity-based lot release system for semiconductor backend assembly environment, *Robot. Comput.-Integrated Manuf.* 23 (2007) 326–338.
- [16] S. French, *Sequencing and Scheduling: An Introduction to the Mathematics of the Job-shop*, Ellis Horwood, Chichester, 1982.
- [17] M. Fu, R. Askin, J. Fowler, M. Haghnevis, N. Keng, J.S. Pettinato, M. Zhang, Batch production scheduling for semiconductor back-end operations, *IEEE Trans. Semicond. Manuf.* 24 (2) (2011) 249–260.
- [18] M. Garey, D. Johnson, *Computers and Intractability: A Guide to the Theory of NP-completeness*, W.H. Freeman, New York, 1979.
- [19] F. Gravetter, L.A. Forzano, *Research methods for the behavioral sciences*, Cengage Learn. (2011).
- [20] S. Jarugumilli, N. Keng, C. Dejong, R. Askin, J. Fowler, Framework for execution level capacity allocation decisions for assembly – test facility using integrated optimization – simulation models, in: *Proceedings of the 2008 Winter Simulation Conference*, 2008, pp. 2292–2297.
- [21] D.J. Johnson, A framework for reducing manufacturing throughput time, *J. Manuf. Syst.* 22 (4) (2003) 283–298.
- [22] A. Klemmt, G. Weigert, An optimization approach for parallel machine problems with dedication constraints: combining simulation and capacity planning, in: *Proceedings of the Winter Simulation Conference*, Winter Simulation Conference, 2011, pp. 1986–1998.
- [23] S.A. Kravchenko, F. Werner, A heuristic algorithm for minimizing mean flow time with unit setups, *Inform. Process. Lett.* 79 (6) (2011) 291–296.
- [24] L.H. Lee, C.H. Chen, E.P. Chew, J. Li, N.A. Pujowidianto, A review of optimal computing budget allocation algorithms for simulation optimization problem, *Int. J. Oper. Res.* 7 (2) (2010) 19–31.
- [25] L.H. Lee, E.P. Chew, P.I. Frazier, Q.S. Jia, C.H. Chen, Advances in simulation optimization and its applications, *IIE Trans.* 45 (7) (2013) 683–684.
- [26] L.H. Lee, E.P. Chew, S. Teng, Y. Chen, Multi-objective simulation-based evolutionary algorithm for an aircraft spare parts allocation problem, *Eur. J. Oper. Res.* 189 (2008) 476–491.

- [27] J. Li, A. Sava, X. Xie, Simulation-based discrete optimization of stochastic discrete event systems subject to non closed-form constraints, *IEEE Trans. Autom. Control* 54 (12) (2009) 2900–2904.
- [28] R.C. Lin, Y. Mustafa, K.S. Pasupathy, Multi-objective simulation optimization using data envelopment analysis and genetic algorithm: specific application to determining optimal resource levels in surgical services, *Omega* 41 (5) (2013) 881–892.
- [29] W. Liu, T.J. Chua, T. Cai, F.Y. Wang, Practical lot release methodology for semiconductor back-end manufacturing, *Prod. Plan. Control* 16 (3) (2005) 297–308.
- [30] L. Monch, J.W. Fowler, S. Dauzere-Peres, S.J. Mason, O. Rose, A survey of problem, solution, and future challenges in scheduling semiconductor manufacturing operations, *J. Sched.* 14 (2011) 583–599.
- [31] D. Nait Tahar, F. Yalaoui, C. Chu, L. Amodeo, A linear programming approach for identical parallel machine scheduling with job splitting and sequence-dependent setup times, *Int. J. Prod. Econ.* 99 (1) (2006) 63–73.
- [32] W.L. Pearn, S.H. Chung, C.M. Lai, Scheduling integrated circuit assembly operations on die bonder, *IEEE Trans. Electron. Packag. Manuf.* 30 (2) (2007) 97–105.
- [33] C. Perkgoz, A. Azaron, H. Katagiri, K. Kato, M. Sakawa, A multi-objective flow time control problem in multi-stage assembly systems using genetic algorithms, *Eur. J. Oper. Res.* 180 (2007) 292–308.
- [34] C.W. Pickardt, T. Hildebrandt, J. Branke, J. Heger, B. Scholz-Reiter, Evolutionary generation of dispatching rule sets for complex dynamic scheduling problems, *Int. J. Prod. Econ.* 145 (1) (2013) 67–77.
- [35] M. Pinedo, *Scheduling: Theory, Algorithms, and Systems*, Prentice-Hall, Englewood Cliffs, NJ, 1995.
- [36] J. Potoradi, O.S. Boon, S.J. Mason, Using simulation-based scheduling to maximize demand fulfillment in a semiconductor assembly facility, in: *Proceedings of 2002 Winter Simulation Conference*, 2002, pp. 1857–1861.
- [37] D. Quadt, H. Kuhn, Production planning in semiconductor assembly, in: *Proceedings of the Fourth Aegean International Conference on Analysis of Manufacturing Systems*, 2003, pp. 181–189.
- [38] I. Ribas, R. Leisten, J.M. Framinan, Review and classification of hybrid flow shop scheduling problems from production system and a solutions procedure perspective, *Comput. Oper. Res.* 37 (2010) 1439–1454.
- [39] R. Ruiz, F.S. Şerifoğlu, T. Urlings, Modeling realistic hybrid flexible flowshop scheduling problems, *Comput. Oper. Res.* 35 (4) (2008) 1151–1175.
- [40] R. Ruiz, J.A. Vázquez-Rodríguez, The hybrid flow shop scheduling problem, *Eur. J. Oper. Res.* 205 (1) (2010) 1–18.
- [41] C. Schmidt, J. Branke, S.E. Chick, Integrating techniques from statistical ranking into evolutionary algorithms, in: *Applications of Evolutionary Computing*, Springer, Berlin Heidelberg, 2006, pp. 752–763.
- [42] A.I. Sivakumar, Multiobjective dynamic scheduling using discrete event simulation, *Int. J. Comput. Integrated Manuf.* 14 (2) (2001) 154–167.
- [43] A.I. Sivakumar, A.K. Gupta, Online multiobjective Pareto optimal dynamic scheduling of semiconductor back-end using conjunctive simulated scheduling, *IEEE Trans. Electron. Packaging Manuf.* 29 (2) (2006) 99–109.
- [44] Y. Sun, C. Zhang, L. Gao, X. Wang, Multi-objective optimization algorithms for flow shop scheduling problem: a review and prospects, *Int. J. Adv. Manuf. Technol.* 55 (2011) 723–739.
- [45] S. Sweat, S. Niu, M.T. Zhang, Z. Zhang, L. Zhang, Multi-factory capacity planning in semiconductor assembly and test manufacturing with multiple-chip products, in: *IEEE International Conference on Automation Science and Engineering*, 2006, pp. 247–252.
- [46] Y. Tang, M. Zhou, R.G. Qiu, Virtual production line design for back-end semiconductor manufacturing system, *IEEE Trans. Semicond. Manuf.* 16 (3) (2003) 543–550.
- [47] E. Tekin, I. Sabuncuoglu, Simulation optimization: a comprehensive review on theory and applications, *IIE Trans.* 36 (2004) 1067–1081.
- [48] E. Vallada, R. Ruiz, A genetic algorithm for the unrelated parallel machine scheduling problem with sequence dependent setup times, *Eur. J. Oper. Res.* 211 (3) (2011) 612–622.
- [49] J.A. Ventura, S.H. Yoon, A new genetic algorithm for lot-streaming flow shop scheduling with limited capacity buffers, *J. Intell. Manuf.* 24 (6) (2013) 1185–1196.
- [50] F.Y. Wang, T.J. Chua, W. Liu, T. Cai, An integrated modeling framework for capacity planning and production scheduling, in: *International Conference on Control and Automation*, 2005, pp. 1137–1162.
- [51] G. Weigert, A. Klemmt, S. Horn, Design and validation of heuristic algorithms for simulation-based scheduling of a semiconductor Backend facility, *Int. J. Prod. Res.* 47 (8) (2009) 2165–2184.
- [52] T. Yang, Y.A. Shen, Heuristic algorithms for a practical-size dynamic parallel-machine scheduling problem: integrated-circuit wire bonding, *Prod. Plan. Control* 23 (1) (2012) 67–78.
- [53] X.F. Yin, T.J. Chun, F.Y. Wang, M.W. Lin, T. Cai, W. Yang, C. Chong, J. Zhu, M. Lam, A rule base heuristic finite capacity scheduling system for semiconductor backend assembly, *Int. J. Comput. Integr. Manuf.* 17 (8) (2004) 733–749.
- [54] M.T. Zhang, S. Niu, S. Deng, Z. Zhang, Q. Li, L. Zheng, Hierarchical capacity planning with reconfigurable kits in global semiconductor assembly and test manufacturing, *IEEE Trans. Autom. Sci. Eng.* 4 (4) (2007) 543–552.